DATASCI507: Proposal

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Abstract

Placeholder

1 Introduction

Can political bulletins tell us something about party's level of centralization?

2 Theory

2.1 Model

The setup goes as follows. A national party P seeks to gain as many votes as possible in order to win an election. To achieve this, it has to convince the local population across the whole country to i) turn out on the day of the election, and ii) vote for party P. To do this, the party has resources that it can allocate across the different constituencies to help promote certain candidates. The party P also needs to decide how *discretionary* its candidates can be in their campaign. For simplicity, we will treat the allocation of resources as fixed for this model, and focus exclusively on the *centralization* choice of party P.

The problem of centralization goes as follows. The leader (L) of party P does not know about the local needs and issues of all the constituencies. Therefore, by granting little discretionary power to the candidates (C) running for its party, he runs the risk of alienating part of the electorate. His priorities might be situated far from those of the local population in constituency d.

On the other hand, the party leader can't delegate all the power and discretionary choices to the candidates. Doing so would be suboptimal, as his preferences might not be aligned with those of all the candidates. Therefore, delegating all of his power to the local candidates runs the risk of leaving these candidates to run on "unwanted platforms". The party still wants to campaign on a relatively similar platform. Therefore, the choice is to decide how far from the party's national platform candidates can run.

The formalization goes as follows. In this game, we have one party leader L and N candidates, one per constituency $d \in \{1, \ldots, N\}$. The party leader L has utility preferences represented

by $u_L(\mathbf{x}, \boldsymbol{\theta}) = -\sum_{d=1}^{N}(x_d - \theta_d)^2$. Here, x_d represents the platform chosen by the candidate in constituency d, and θ_d is the optimal point of constituency d. Each constituency's optimal point is drawn i.i.d. from a distribution F, $\theta_d \sim F$. The party leader doesn't observe the realization of θ_d, but he does know the distribution F.

The candidate in constituency d has utility preferences represented by $u_{C_d}(x_d, \theta_d, b) = -(x_d - \theta_d - b)^2$, where x_d is the candidate's platform, θ_d is the local optimal point, and b is the bias, or disagreement, of the candidates with the party leader, assumed common to all candidates. Each candidate gets to observe θ_d before choosing their own platform x_d.

The game follows this timing: L commits to a set of permitted platforms $\mathcal{D}$; Nature draws θ_d; candidate d observes θ_d; candidate d chooses $x_d \in \mathcal{D}$.

2.2 Literature

3 Data

This project collects an original dataset using political bulletins in Japan for the 2024, 2025, and 2026 lower house elections. The pipeline goes as follow. We first developped a webscrapper to retrieve the election bulletins from go2senkyo.com. The website compiles all the election bulletins across all districts retrieved from each prefecture's electoral commission. The scrapper gets all the PDFs files and store them locally. Once the files are downloaded, another script uses Claude API to segment all the PDFs into the candidates running inside district d. This approach is to facilitate the following step, which is to use PDF parsing and OCR, and to increase the precision of the information retrieved. About the parsing and the OCR, we used the Yomitoku Python librairy that is optimized for Japanese text and documents.

Once the parsing is done, we can store the information into a Pandas dataframe. Each candidates (rows) have the following information attached to them: 1) text from their election bulletin, 2) the number of paragraphs, 3) the number of headings, 4) the total number of characters (including both Kanji and Kana).

Figure 1 present an example of an electoral bulletin for one candidate.

Figure 2 presents the pipeline of the project, and the future extension that can be made from the data generated.



Figure 1: Political candidate in Aichi prefecture

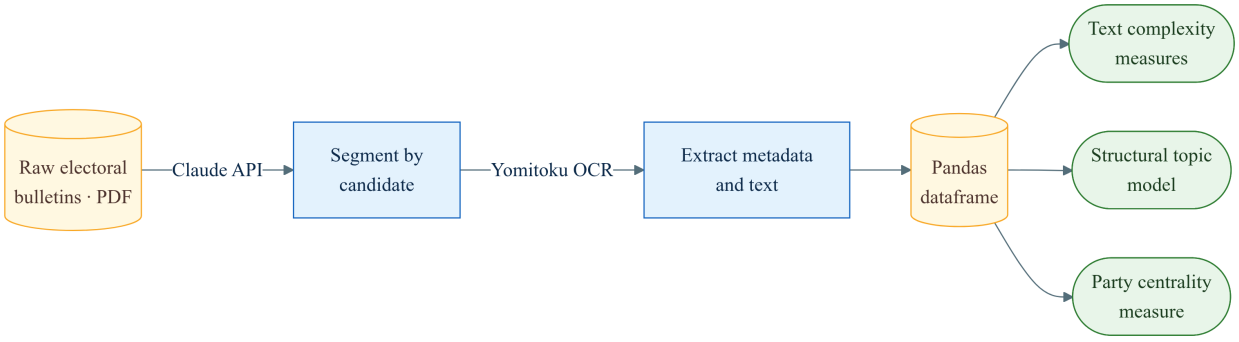


Figure 2: Pipeline of the project and future extension